

Constitutional Megaloblastic Anemia With Severe Neurologic Disease (DHFR Deficiency)

Disease knowledge-base research report MONDO:0013456 · OMIM #613839 · Orphanet
ORPHA:319651 · Gene: DHFR Date: 2026-09-16

Summary (Answer to the Research Question)

Constitutional Megaloblastic Anemia With Severe Neurologic Disease is an ultra-rare **autosomal recessive inborn error of folate metabolism caused by biallelic loss-of-function missense mutations in *DHFR*** (dihydrofolate reductase; chromosome 5q14.1). Profound DHFR enzyme deficiency prevents regeneration of tetrahydrofolate (THF) from dihydrofolate and the reduction of dietary folate, collapsing the reduced-folate pool. The result is a dual hematologic + neurologic disease: **megaloblastic anemia/pancytopenia** (impaired thymidylate and purine synthesis) plus **severe cerebral folate deficiency** (low CSF 5-methyltetrahydrofolate) and **cerebral tetrahydrobiopterin (BH4) deficiency**, manifesting as developmental delay, seizures (notably atypical childhood absence epilepsy), and other neurologic abnormalities. The disease is **treatable with folinic acid (calcium leucovorin, a pre-reduced folate that bypasses the DHFR block)**, which corrects the anemia, normalizes CSF folate, and improves neurologic symptoms — while folic acid is ineffective because it still requires DHFR for activation.

The disease was defined by two simultaneous 2011 reports (Cario et al.,

[P 21310277](#) Banka et al.,

[P 21310276](#)) in a total of 6 patients from 4 consanguineous/related families. Because reported cases are very few, much of the clinical detail below derives from these primary reports and a subsequent folate-disorder review (Watkins & Rosenblatt,

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Evidence source types are flagged throughout: [HUMAN clinical], [in vitro], [structural/biochemical], [review].

1. Disease Information

- **Overview:** A recessive inborn error of intracellular folate metabolism combining bone-marrow failure (megaloblastic anemia/pancytopenia) with a severe central-nervous-system phenotype driven by cerebral folate and BH4 deficiency. It is a **treatable metabolic encephalopathy** when recognized early. [HUMAN clinical;

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- **Key identifiers:**
- **MONDO:** MONDO:0013456
- **OMIM (phenotype):** #613839 "Megaloblastic anemia due to dihydrofolate reductase deficiency"
- **OMIM (gene):** *126060 (DHFR)
- **Orphanet:** ORPHA:319651 (Megaloblastic anemia due to dihydrofolate reductase deficiency)
- **Gene / HGNC:** DHFR (HGNC:2861); NCBI Gene 1719; Ensembl ENSG00000228716; UniProt P00374
- **ICD-10:** D53.1 (Other megaloblastic anemias, not elsewhere classified); **ICD-11:** 3A01.0 area / 5C50 (inborn errors of metabolism) — no unique code
- **MeSH:** Folic Acid Deficiency (D005494); Anemia, Megaloblastic (D000749); Tetrahydrofolate Dehydrogenase (D013894)
- **Synonyms / alternative names:** DHFR deficiency; Dihydrofolate reductase deficiency; Megaloblastic anemia due to dihydrofolate reductase deficiency; Constitutional megaloblastic anemia with severe neurologic disease; Cerebral folate deficiency due to DHFR deficiency.

- **Information source type:** Aggregated disease-level (OMIM/Orphanet/MONDO) plus **individual-patient** case series (EHR-derived clinical descriptions of ~6 patients). Not derived from large population EHR datasets.

2. Etiology

- **Primary cause (genetic):** Biallelic (homozygous) germline **missense mutations in *DHFR*** producing a stable but catalytically deficient enzyme. Cario et al. identified **c.458A>T (p.Asp153Val)**; Banka et al. identified a homozygous *DHFR* missense mutation resulting in profound enzyme deficiency. [HUMAN clinical;

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"DHFR sequencing revealed a homozygous DHFR mutation, c.458A>T (p.Asp153Val), in all siblings."

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- **Genetic risk factors:** Homozygous DHFR pathogenic variants are causal (Mendelian, not susceptibility loci). **Consanguinity / relatedness** is a major risk factor — reported families were consanguineous or distantly related, consistent with a rare recessive founder-type allele.
- **Environmental risk factors:** None are causal. However, **low dietary folate intake would be expected to worsen the phenotype**, and **exposure to antifolate drugs** (methotrexate, trimethoprim, pyrimethamine) that further inhibit residual DHFR could exacerbate disease (mechanistic inference).
- **Protective factors:** **Dietary folate as reduced folates (folinic acid / 5-formyl-THF, 5-MTHF)** bypasses the block and is protective/therapeutic. Common population *DHFR* polymorphisms (e.g., the 19-bp intron-1 deletion, c.594+59del19; p.Leu80Phe) modulate folate handling/antifolate response in the general population but are **not** causes of this monogenic disease.

- **Gene–environment interactions:** Folate/antifolate status interacts strongly with residual DHFR activity — the phenotype is a genetic enzyme deficiency whose severity is modifiable by folate form and dose (folinic acid rescue is the clearest example). [HUMAN clinical;

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3. Phenotypes

Frequencies are qualitative given the very small case number ($n \approx 6$). Onset is typically **neonatal to infancy/early childhood**; course is **progressive if untreated, largely reversible/stabilizable with folinic acid**.

Hematologic (laboratory abnormalities / clinical signs): - **Megaloblastic anemia** — core feature, most/all patients. HPO: **HP:0001889** (Megaloblastic anemia). Onset infancy; severe; treatment-responsive. - **Pancytopenia** — reported in some patients. HP:0001876 (Pancytopenia); HP:0001873 (Thrombocytopenia), HP:0001882 (Leukopenia). - **Macrocytosis / elevated MCV**, megaloblastic bone marrow, hypersegmented neutrophils. HP:0001972 (Macrocytic anemia).

"characterized by megaloblastic anemia and/or pancytopenia"
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Neurologic (symptoms / signs): - **Seizures / epilepsy**, characteristically **atypical childhood absence epilepsy**. HP:0002121 (Absence seizure), HP:0001250 (Seizure). [HUMAN clinical;

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"megaloblastic anemia and cerebral folate deficiency causing neurologic disease with atypical childhood absence epilepsy."
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- **Global developmental delay / intellectual disability**. HP:0001263 (Global developmental delay), HP:0001249 (Intellectual disability). - **Microcephaly** (reported in

cerebral folate deficiency states). HP:0000252. - **Cerebral folate deficiency features** — variable neurologic findings including hypotonia, movement/motor abnormalities, and, mechanistically expected from BH4/monoamine deficiency, potential extrapyramidal or mood/behavioral changes. HP:0002376 (Developmental regression) variably. - **Neuroimaging abnormalities** consistent with folate-deficient leukoencephalopathy in some patients. HP:0002352 (Leukoencephalopathy).

Severity / progression / QoL: Severe, potentially life-threatening in the neonatal/infantile period (anemia) with substantial neurodisability risk. **Quality-of-life impact is high if untreated** (epilepsy + developmental impairment); **early folinic acid markedly improves hematologic status and neurologic trajectory**, though pre-treatment CNS injury may persist. [HUMAN clinical;

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4. Genetic / Molecular Information

- **Causal gene: DHFR** (dihydrofolate reductase), 5q14.1; OMIM *126060; HGNC:2861; UniProt P00374 (187 aa cytosolic enzyme). EC 1.5.1.3.
- **Pathogenic variants (germline, autosomal recessive):**
- **c.458A>T; p.Asp153Val (p.D153V)** — homozygous; 3 affected siblings (Cario et al.). Missense; classified pathogenic; drastically reduced enzyme activity and reduced methotrexate (FMTX) binding; **normal mRNA but reduced protein → destabilizing missense / loss of function.** [HUMAN clinical + in vitro;

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> "RT-PCR of DHFR mRNA revealed no differences between wild-type and DHFR mutation-carrying cells, whereas protein expression was reduced in cells with the DHFR mutation."

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- **Homozygous DHFR missense mutation** in 3 individuals from 2 families (Banka et al.) → profound enzyme deficiency (reported as p.Leu80Phe in that cohort). [HUMAN clinical;

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- **Variant type/class:** missense (loss-of-function via reduced protein stability/activity). No nonsense/frameshift/structural variants reported as causal to date.
- **ACMG/AMP:** Pathogenic (functional enzyme assays + segregation + rarity).
- **Allele frequency:** Causal alleles are private/ultra-rare (essentially absent in gnomAD as homozygotes). Note the **common benign DHFR** 19-bp intron-1 deletion and p.Leu80Phe polymorphism exist in general populations and affect folate/antifolate pharmacology but are not disease-causing here.
- **Somatic vs germline:** Germline only.
- **Functional consequence: Loss of function** (reduced catalytic activity + reduced protein). Heterozygotes show **intermediate** DHFR activity and FMTX binding but are clinically unaffected (recessive). [in vitro;

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- **Modifier genes:** Not formally established; folate-pathway genes (MTHFR, folate transporters SLC46A1/FOLR1, MTHFD1) and dietary folate plausibly modify severity (inference).
- **Epigenetic / chromosomal abnormalities:** None implicated; no aneuploidy/translocation. Global one-carbon/methylation supply is indirectly reduced (methionine/SAM), a downstream metabolic — not primary epigenetic — effect.

5. Environmental Information

- **Environmental factors:** No environmental cause. **Antifolate drug exposure** (methotrexate, trimethoprim–sulfamethoxazole, pyrimethamine) is mechanistically contraindicated/aggravating because it further inhibits DHFR (inference).
- **Lifestyle / diet: Folate nutrition** is the dominant modifiable factor; adequacy of *reduced* folate (folinic acid) is protective. Ordinary folic acid supplementation does not rescue the defect. [HUMAN clinical;

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- **Infectious agents:** Not applicable (not an infectious disease).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *DHFR* missense mutation** (e.g., p.Asp153Val) → **destabilized DHFR protein with severely reduced catalytic activity** ([in vitro] reduced enzyme activity + FMTX binding; normal mRNA, low protein). *Demonstrated*.
 2. Reduced DHFR activity → **failure to reduce 7,8-dihydrofolate (DHF) to tetrahydrofolate (THF)** and failure to reduce dietary folic acid → **depletion of the reduced-folate (THF) pool**. *Demonstrated by patient folate profiling*.
 3. THF depletion **branches** into three arms:
 4. **3a. Hematologic arm:** low THF → impaired **thymidylate synthase cycle** (dUMP→dTMP) and de novo purine synthesis → **uracil misincorporation / imbalanced dNTPs** → **ineffective DNA synthesis in erythroid precursors** → nuclear–cytoplasmic asynchrony → **megaloblastic anemia and pancytopenia**. *Inferred from folate biochemistry; demonstrated hematologic phenotype*.
 5. **3b. Cerebral folate arm:** impaired regeneration/transport of reduced folate → **low CSF 5-methyltetrahydrofolate (cerebral folate deficiency)** → impaired CNS one-carbon metabolism, methylation, and neurotransmitter precursor supply → **seizures (atypical absence epilepsy), developmental delay**. *Demonstrated (low CSF folate)*.
 6. **3c. Neurotransmitter/BH4 arm:** DHFR also **regenerates tetrahydrobiopterin (BH4) from dihydrobiopterin (BH2)** (the salvage arm of BH4 metabolism). Loss → **cerebral BH4 deficiency** → reduced activity of tyrosine/tryptophan/phenylalanine hydroxylases → **reduced dopamine, serotonin, norepinephrine** → neurologic/neuropsychiatric manifestations. *Demonstrated (cerebral BH4 deficiency;*
- P** 21310276
7. Convergent CNS effects of 3b + 3c → **severe, potentially progressive encephalopathy**.
 8. **Therapeutic branch (reversal):** administration of **folinic acid (5-formyl-THF)**, a folate already reduced beyond the DHFR block → restores THF pool → **corrects anemia, normalizes CSF folate, improves neurologic symptoms**. *Demonstrated*.

Detail by category

- **Molecular pathways:** Folate one-carbon metabolism (KEGG hsa00670 one-carbon pool by folate; Reactome "Metabolism of folate and pterines"); pterine/BH4 salvage; methionine cycle (SAM methylation).
- **Cellular processes:** DNA replication in rapidly dividing cells (erythroblasts) → ineffective erythropoiesis/apoptosis of precursors; neuronal signaling via monoamines.
- **Protein dysfunction:** Destabilizing missense → **loss of function** with reduced protein abundance (not simply catalytic-site abolition); heterozygous intermediate activity. DHFR is a small (~21 kDa, 187 aa) NADPH-dependent oxidoreductase (Rossmann-like fold) whose catalysis involves hydride transfer from NADPH and protonation of the folate N5 — mechanistic basis for why point mutations near substrate/cofactor contacts abolish activity. [structural;

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Human DHFR crystal structures (e.g., PDB via Cody et al.,

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define an active-site pocket with a conserved substrate-anchoring **Arg70**, and pteridine-binding residues **Phe31**, **Gln35**, **Val115** (Val115 vs Ile in microbial DHFR underlies antifolate selectivity). The disease missense residues (e.g., Asp153, Leu80) map to this compact fold; substitutions destabilize the protein and/or perturb cofactor/substrate binding, consistent with the observed loss of activity and reduced protein level. UniProt P00374; AlphaFold model AF-P00374.

- **Metabolic changes:** ↓ THE, ↑ DHF (relative), ↓ 5-MTHF (esp. CSF), ↓ BH4, impaired dTMP/ purine synthesis, impaired homocysteine remethylation (variable). CHEBI: 7,8-dihydrofolate (CHEBI:20506), tetrahydrofolate (CHEBI:26907), 5-methyltetrahydrofolate (CHEBI:15641), folinic acid (CHEBI:63606), tetrahydrobiopterin (CHEBI:15372).
- **Immune involvement:** Not a primary feature (contrast MTHFD1 deficiency, which adds SCID).
- **Tissue damage mechanisms:** Ineffective hematopoiesis (marrow); neuronal dysfunction from substrate/neurotransmitter insufficiency (largely functional, potentially structural leukoencephalopathy).
- **Biochemical abnormality:** Enzyme deficiency — dihydrofolate reductase (EC 1.5.1.3).

- **Suggested GO terms:** GO:0046452 (dihydrofolate metabolic process), GO:0006545 (glycine biosynthetic process), GO:0046655 (folic acid metabolic process), GO:0009394 (2'-deoxyribonucleotide metabolic process), GO:0006760 (folic acid-containing compound metabolic process), GO:0034355 (NAD salvage n/a) ; molecular function GO:0004146 (dihydrofolate reductase activity); GO:0006559 (l-phenylalanine catabolic process, via BH4).
 - **Suggested CL terms:** erythroid progenitor cell (CL:0000038), megakaryocyte-erythroid progenitor, neuron (CL:0000540), dopaminergic neuron (CL:0000700), serotonergic neuron (CL:0000850).
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7. Anatomical Structures Affected

- **Organ level (primary):** Bone marrow / hematopoietic system (UBERON:0002371 bone marrow) and brain / central nervous system (UBERON:0000955 brain).
 - **Secondary/system involvement:** Peripheral blood (all lineages — anemia, leukopenia, thrombocytopenia); potentially cerebrospinal fluid compartment (UBERON:0001359 CSF) showing low 5-MTHF.
 - **Body systems:** Hematologic/immune (blood) and nervous system; digestive absorption of folate is intact (the defect is intracellular reduction, distinguishing it from hereditary folate malabsorption).
 - **Tissue/cell level:** Erythroid and other hematopoietic precursors (megaloblastic marrow); CNS neurons dependent on monoamine neurotransmitters. CL: erythroid progenitor (CL:0000038), neuron (CL:0000540).
 - **Subcellular level: Cytosol** (DHFR is cytosolic; GO:0005829) with folate-dependent one-carbon reactions in cytosol and nucleus; downstream **nuclear** de novo thymidylate synthesis affected.
 - **Localization / lateralization:** Systemic/bilateral; CNS involvement is diffuse/bilateral (e.g., diffuse white-matter changes), not focal.
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8. Temporal Development

- **Onset:** Congenital enzyme defect; clinical presentation typically **neonatal to early infancy/childhood** (megaloblastic anemia early; seizures/absence epilepsy in early childhood).

[HUMAN clinical;

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- **Onset pattern:** Subacute/chronic, insidious neurologic decline with intercurrent anemia.
- **Progression: Progressive if untreated;** neurologic damage may accrue during the untreated window. With folinic acid, hematologic and biochemical parameters normalize and neurologic symptoms improve/stabilize. [HUMAN clinical;

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- **Course pattern:** Chronic, lifelong (requires lifelong folinic acid). Seizures may be episodic.
- **Remission: Treatment-induced** biochemical/hematologic remission with folinic acid; not spontaneous.
- **Critical period: Early infancy/childhood is the critical intervention window** — earlier treatment better preserves neurodevelopment (rationale for newborn/early metabolic detection).

9. Inheritance and Population

- **Epidemiology: Ultra-rare** — only a handful of patients (≈ 6 from 4 families) reported worldwide; prevalence $< 1/1,000,000$ (Orphanet "unknown/ultra-rare"). Incidence not quantifiable.
- **Inheritance: Autosomal recessive.** Heterozygous carriers are asymptomatic with intermediate enzyme activity. [in vitro;

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- **Penetrance:** Complete in biallelic individuals reported; **expressivity variable** (neurologic severity varies — "variable neurological findings,"

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- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.
- **Founder effects / consanguinity:** Reported families were **consanguineous or distantly related**; private homozygous alleles — consistent with founder/consanguinity mechanism.
- **Carrier frequency:** Unknown; expected very low given rarity.

- **Demographics:** No established ethnic predilection given tiny sample; both sexes affected (autosomal). No sex bias expected. Age distribution: pediatric onset.

10. Diagnostics

- **Laboratory tests:**
- CBC + smear: **macrocytic/megaloblastic anemia**, possible pancytopenia, hypersegmented neutrophils. LOINC: MCV 30428-7, Hemoglobin 718-7.
- Bone marrow: megaloblastic changes.
- **Serum/RBC folate, vitamin B12, homocysteine** to position within folate disorders (B12 normal; distinguishes from B12 deficiency).
- **CSF 5-methyltetrahydrofolate (low)** — hallmark of cerebral folate deficiency. [HUMAN clinical;

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- **CSF neurotransmitter metabolites / biopterins** (low BH4, altered HVA/5-HIAA) reflecting BH4/monoamine deficiency. [HUMAN clinical;

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- LC-MS/MS folate profiling of RBC, plasma, CSF (as used diagnostically). [PMID 21310277]
- **Functional/enzyme assay: DHFR enzyme activity and fluorescein-methotrexate (FMTX) binding** in lymphoblastoid cells/fibroblasts — severely reduced in patients, intermediate in carriers. [in vitro;

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- **Imaging:** Brain MRI — may show leukoencephalopathy/white-matter changes; used to characterize CNS involvement.
- **Electrophysiology:** EEG for absence/atypical absence epilepsy characterization.
- **Genetic testing: Definitive test = DHFR sequencing** (single-gene, or as part of a megaloblastic-anemia / inborn-errors-of-folate / epilepsy-metabolic gene panel; WES/WGS in undiagnosed cases). Homozygosity mapping was pivotal in gene discovery. [PMID 21310277]. GTR panels for "megaloblastic anemia" and "cerebral folate deficiency" include DHFR.

- **Clinical criteria / differential diagnosis:** Combination of megaloblastic anemia + low CSF folate + folinic-acid responsiveness + biallelic DHFR variants. **Differentiate from:** vitamin B12 deficiency; folate malabsorption (SLC46A1); FOLR1 cerebral folate deficiency (no anemia); MTHFR deficiency (homocystinuria, no megaloblastic anemia); **MTHFD1 deficiency (adds SCID/immunodeficiency; impaired nuclear dTMP synthesis with elevated uracil in DNA,**

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thiamine-responsive megaloblastic anemia (SLC19A2); orotic aciduria.

- **Screening:** No routine newborn screen currently; consider **cascade/carrier testing** in affected families and prenatal/preimplantation testing where the familial variant is known.

11. Outcome / Prognosis

- **Survival/mortality:** Untreated severe megaloblastic anemia/pancytopenia is potentially fatal in infancy; **prognosis is substantially improved with early folinic acid.** No formal survival statistics exist (too few cases).
- **Morbidity/function:** Neurodevelopmental disability and epilepsy are the main morbidities; degree depends on timing of treatment. Hematologic parameters are fully correctable.
- **Recovery potential: Hematologic and biochemical abnormalities are reversible with folinic acid; neurologic recovery is partial** — pre-treatment CNS injury may be irreversible, underscoring early diagnosis. [HUMAN clinical;

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The analogous FOLR1 cerebral folate deficiency literature reinforces a **critical treatment window:** folinic acid begun after >15 years of illness produced no meaningful clinical/neurophysiological improvement in three siblings

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whereas timely initiation improved outcome

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- **Prognostic factors:** Age at diagnosis/treatment initiation, extent of pre-treatment neurologic damage, adherence to lifelong folinic acid, avoidance of antifolate drugs.

12. Treatment

- **First-line pharmacotherapy: Folinic acid (calcium leucovorin / 5-formyltetrahydrofolate)** — a pre-reduced folate that bypasses the DHFR block; corrects anemia/pancytopenia, normalizes CSF folate, improves neurologic symptoms. Lifelong. NCIT: **Leucovorin Calcium (C1035) / Folinic Acid**. [HUMAN clinical;

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"Treatment with folinic acid resulted in the resolution of hematological abnormalities, normalization of CSF folate levels, and improvement of neurological symptoms."

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- **Ineffective/avoid: Folic acid** (requires DHFR to become active — does not rescue). Beyond needing DHFR, **unmetabolized folic acid actively inhibits 5-MTHF transport across the blood–CSF barrier**, worsening cerebral folate deficiency: in two CFD cases high-dose folic acid failed to normalize CSF 5-MTHF, and stopping folic acid normalized it — so folinic acid or 5-MTHF is preferred

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Avoid **antifolate drugs** (methotrexate, trimethoprim, pyrimethamine).

"In the treatment of CFD, supplementation of folinic acid or 5MTHF (in cases of impaired 5MTHF synthesis) is preferred over the use of FA."

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- **Adjunctive:** Dose titration to normalize CSF 5-MTHF; consider that some cerebral folate deficiency benefits from higher folinic acid doses to cross the blood–brain barrier.

Antiepileptic therapy for seizures as needed; monitor since folate status interacts with some AEDs.

- **Potential neurotransmitter support:** Because of **BH4/monoamine deficiency**, consideration of BH4 (sapropterin) or neurotransmitter precursor supplementation (e.g., L-dopa/carbidopa, 5-hydroxytryptophan) is biologically rational in select patients (inference; not established as standard for DHFR deficiency). Management principles for BH4/monoamine-deficiency disorders are summarized in an international consensus guideline (

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- **Advanced/experimental therapeutics:** No gene, cell, or RNA therapy in clinical use; the disorder is largely managed by metabolite replacement. Gene replacement is conceptually feasible but unstudied.
- **Supportive care:** Transfusion for severe anemia acutely; developmental/rehabilitative therapies for neurodisability.
- **Treatment strategy / personalized medicine:** Genotype-confirmed diagnosis → lifelong folinic acid + monitoring of CBC, plasma and CSF folate; family cascade testing. NCIT terms: Leucovorin Calcium (C1035); Sapropterin Dihydrochloride (C61815) [experimental rationale].

13. Prevention

- **Primary prevention:** Not preventable in an affected homozygote (germline). **Genetic counseling** for at-risk families; **carrier/cascade testing; prenatal or preimplantation genetic testing** where the familial DHFR variant is known.
- **Secondary prevention (early detection):** High index of suspicion in infants with megaloblastic anemia + neurologic signs + low CSF folate → early DHFR testing and prompt folinic acid to prevent CNS damage. No population newborn screen exists yet, but the treatable nature makes early metabolic detection valuable.
- **Tertiary prevention:** Lifelong folinic acid, avoidance of antifolates, seizure control, developmental support to prevent complications and progression.
- **Counseling:** Autosomal recessive recurrence risk 25% per pregnancy for carrier couples; offer NSGC/ACMG-guided counseling.

14. Other Species / Natural Disease

- **Taxonomy:** DHFR is universally conserved (essential enzyme) across bacteria, fungi, plants, and animals (NCBI Taxon relevance broad). Human host: *Homo sapiens* (NCBI:txid9606).
- **Orthologous genes:** Mouse *Dhfr* (NCBI Gene 13361); rat *Dhfr* (NCBI Gene 24312); zebrafish *dhfr*. High evolutionary conservation of catalytic function.
- **Natural disease in animals:** No well-characterized spontaneous DHFR-deficiency disease reported in companion animals/wildlife (OMIA — none established). DHFR is a classic antimicrobial/antineoplastic drug target across species.
- **Comparative biology:** Complete DHFR loss is embryonic-lethal in model organisms (essential gene), which is why human disease arises only from **hypomorphic missense** alleles retaining partial activity — a key comparative insight.
- **Transmission:** Not applicable (non-infectious, monogenic).

15. Model Organisms

- **Model status:** No published dedicated mouse/zebrafish model that recapitulates the *human hypomorphic DHFR-deficiency disease* was identified in this search; complete *Dhfr* knockouts are expected to be embryonic-lethal (essential enzyme), a key limitation.
- **Cellular/in vitro models (used in the defining studies): Patient-derived EBV-immortalized lymphoblastoid cell lines and fibroblasts** demonstrating reduced DHFR activity, reduced FMTX binding, and reduced protein — the primary functional models to date. [in vitro;

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- **Model types feasible:** Knock-in of patient missense alleles (e.g., p.Asp153Val, p.Leu80Phe) in mouse/human iPSC or cell lines; conditional/hypomorphic alleles; iPSC-derived neurons/erythroid cultures to model the dual phenotype; humanized DHFR yeast/bacterial complementation assays for variant functional testing.
- **Applications:** Variant functional classification (enzyme activity, antifolate binding), folate-flux studies, testing folinic acid / BH4 rescue.
- **Resources:** MGI (*Dhfr*), Cellosaurus (patient lines), ClinVar (DHFR variants).

Ontology Term Quick-Reference

- **Disease:** MONDO:0013456; OMIM #613839; ORPHA:319651
 - **Gene/Protein:** DHFR (HGNC:2861; NCBI 1719; UniProt P00374); GO:0004146 (DHFR activity)
 - **HPO:** HP:0001889 (megaloblastic anemia), HP:0001876 (pancytopenia), HP:0002121 (absence seizure), HP:0001250 (seizure), HP:0001263 (global developmental delay), HP:0001249 (intellectual disability), HP:0000252 (microcephaly), HP:0002352 (leukoencephalopathy)
 - **GO (BP/CC):** GO:0046655 (folic acid metabolic process), GO:0046452 (dihydrofolate metabolic process), GO:0006760 (folic acid-containing compound metabolic process), GO:0005829 (cytosol)
 - **CL:** CL:0000038 (erythroid progenitor), CL:0000540 (neuron), CL:0000700 (dopaminergic neuron)
 - **UBERON:** UBERON:0002371 (bone marrow), UBERON:0000955 (brain), UBERON:0001359 (CSF)
 - **CHEBI:** CHEBI:20506 (7,8-dihydrofolate), CHEBI:26907 (tetrahydrofolate), CHEBI:15641 (5-MTHF), CHEBI:63606 (folinic acid), CHEBI:15372 (tetrahydrobiopterin)
 - **NCIT:** Leucovorin Calcium (C1035); Sapropterin (C61815)
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Supported vs. Refuted Hypotheses

Supported: - H1 (Supported): Biallelic *DHFR* missense mutations cause the disease via profound enzyme loss of function. [PMID 21310277, 21310276] - H2 (Supported): The mechanism is THF-pool depletion producing megaloblastic anemia + cerebral folate deficiency + cerebral BH4 deficiency. [PMID 21310276, 21310277] - H3 (Supported): Folinic acid (not folic acid) is corrective because it bypasses the DHFR block. [PMID 21310277, 21310276] - H4 (Supported): The disorder is distinguishable within inborn errors of folate metabolism; MTHFD1 deficiency adds immunodeficiency. [PMID 22108709, 25548164]

Refuted / Not supported: - Environmental or infectious primary etiology — refuted (Mendelian recessive enzyme defect). - Dominant inheritance — refuted (heterozygotes asymptomatic with intermediate activity).

Limitations and Future Directions

- **Evidence base is very small** (~6 patients from 4 families); frequencies, prognosis, and genotype–phenotype correlations are qualitative. Numbers such as prevalence are order-of-magnitude estimates from Orphanet-class sources.
 - The curated literature accessible here is limited; additional post-2012 case reports and any dedicated animal models were not retrievable in this environment and should be sought in OMIM/GeneReviews/HGMD/ClinVar and full PubMed.
 - **Future directions:** systematic natural-history and treatment-timing studies; standardized CSF 5-MTHF/BH4/neurotransmitter panels; functional assays for variant classification; evaluation of BH4/neurotransmitter-precursor adjuncts; and consideration of DHFR in metabolic/epilepsy newborn or early-childhood screening given treatability.
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Primary References (PMID)

- **21310277** — Cario H, et al. *Dihydrofolate reductase deficiency due to a homozygous DHFR mutation causes megaloblastic anemia and cerebral folate deficiency leading to severe neurologic disease.* Am J Hum Genet, 2011. [HUMAN clinical + in vitro]
- **21310276** — Banka S, et al. *Identification and characterization of an inborn error of metabolism caused by dihydrofolate reductase deficiency.* Am J Hum Genet, 2011. [HUMAN clinical + in vitro]
- **22108709** — Watkins D, Rosenblatt DS. *Update and new concepts in vitamin responsive disorders of folate transport and metabolism.* J Inherit Metab Dis, 2012. [review]
- **25548164** — Field MS, et al. *Human mutations in MTHFD1 impair nuclear de novo thymidylate biosynthesis.* (differential) [HUMAN + in vitro]
- **25453083** — Wan Q, et al. *Toward resolving the catalytic mechanism of dihydrofolate reductase using neutron and ultrahigh-resolution X-ray crystallography.* [structural — DHFR enzymology]
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